Homework 9

Week 11

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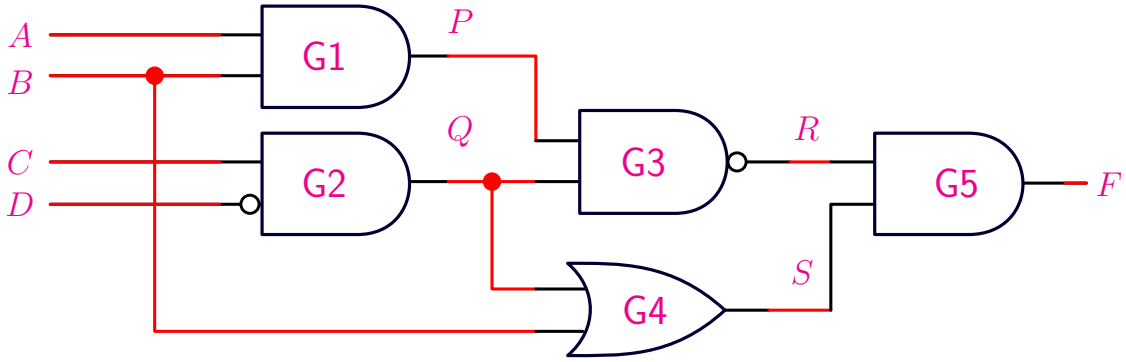
1 Delay Fault Testing

From the circuit below, Find Delay Fault Test Input Vector in case of



Circuit functions (used in both problems):

$$P = AB, \qquad Q = C\overline{D}, \qquad R = \overline{PQ}, \qquad S = Q + B,$$
$$F = R \cdot S = \overline{AB\,C\overline{D}} \cdot \left(C\overline{D} + B\right).$$

A delay fault test is a vector pair $\langle V_1, V_2 \rangle$ (written as $ABCD$): V_1 initializes the path, V_2 launches the transition. To sensitize a path, every side input must be held *stable* at the non-controlling value of its gate (1 for AND/NAND, 0 for OR) in both vectors, so that the output transition depends only on the transition travelling along the path (robust test).

Problem 1. $\downarrow A$ transition

Solution

Path: $A \to P\,(G1) \to R\,(G3) \to F\,(G5)$.

- **G1 (AND):** side input $B = 1$.
- **G3 (NAND):** side input $Q = C\overline{D} = 1 \Rightarrow C = 1,\ D = 0$.
- **G5 (AND):** side input $S = Q + B = 1$ (already satisfied, and stable since $Q = 1$).

	A	B	C	D	P	Q	R	S	F
V_1	1	1	1	0	1	1	0	1	0
V_2	0	1	1	0	0	1	1	1	1
	$\downarrow$	1	1	0	$\downarrow$	1	$\uparrow$	1	$\uparrow$

$$\boxed{\langle V_1, V_2 \rangle = \langle ABCD = 1110,\ ABCD = 0110 \rangle}$$

The $\downarrow A$ transition appears at the output as $\uparrow F$ (inverted once by the NAND). If F is sampled before it has risen to 1, the path A-P-R-F has a delay fault. This test is unique, since B, C, D are all forced.

Problem 2. $\uparrow B$ transition

Solution

B fans out to two paths that reconverge at G5, so each path needs its own test:

$$\text{Path 1: } B \to P\,(G1) \to R\,(G3) \to F\,(G5), \qquad \text{Path 2: } B \to S\,(G4) \to F\,(G5).$$

Path 1 (B-P-R-F):

- **G1 (AND):** $A = 1$.
- **G3 (NAND):** $Q = 1 \Rightarrow C = 1,\ D = 0$.
- **G5 (AND):** $S = Q + B = 1$, stable because $Q = 1$ holds S at 1 while B changes.

	A	B	C	D	P	Q	R	S	F
V_1	1	0	1	0	0	1	1	1	1
V_2	1	1	1	0	1	1	0	1	0
	1	$\uparrow$	1	0	$\uparrow$	1	$\downarrow$	1	$\downarrow$

Path 1: $\langle V_1, V_2\rangle = \langle 1010,\ 1110\rangle$ $\quad (\uparrow B \Rightarrow \downarrow F)$

Path 2 (B-S-F):

- **G4 (OR):** $Q = C\overline{D} = 0 \Rightarrow (C, D) \neq (1, 0)$.
- **G5 (AND):** $R = \overline{PQ} = 1$, which holds automatically (and stays stable) because $Q = 0$.
- A is a don't-care; choose $A = 0$, $C = D = 0$.

	A	B	C	D	P	Q	R	S	F
V_1	0	0	0	0	0	0	1	0	0
V_2	0	1	0	0	0	0	1	1	1
	0	$\uparrow$	0	0	0	0	1	$\uparrow$	$\uparrow$

Path 2: $\langle V_1, V_2\rangle = \langle 0000,\ 0100\rangle$ $\quad (\uparrow B \Rightarrow \uparrow F)$

(Any $A \in \{0, 1\}$ with $CD \in \{00, 01, 11\}$ also works, e.g. $\langle 1001, 1101\rangle$.)

The two paths cannot be tested by one vector pair: Path 1 requires $Q = 1$ while Path 2 requires $Q = 0$. Hence both pairs are needed to cover $\uparrow B$.